

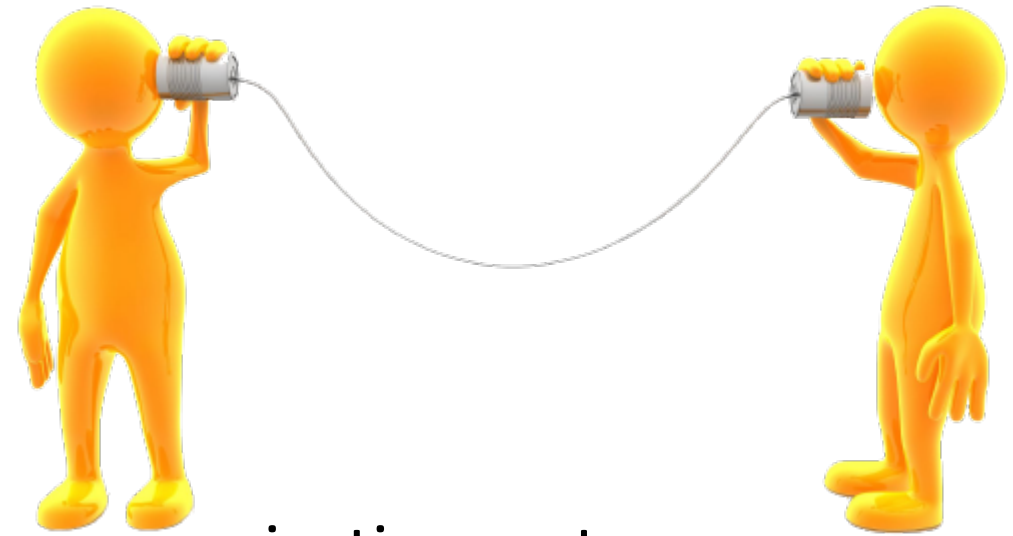
Lecture 7

Machine-to-Machine (M2M)

Communication

- What is communication
- Interfaces
- Protocol
- Synchronous and Asynchronous communication
- The OSI model (communication is divided into layers)
- Topology and communication principles
- Handshake signals (Strobe)
- I/O communication
- Introducing Field bus

Communication

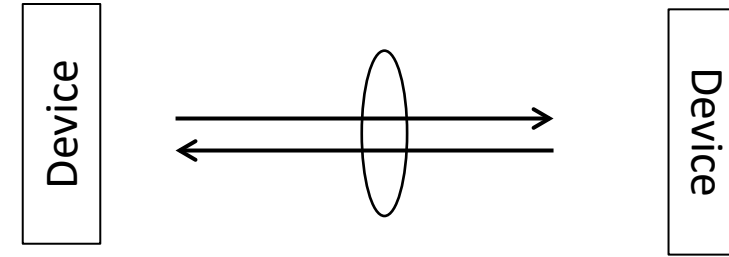


- Common rules and standards in the communication system are necessary to avoid misunderstandings and error
 - Several communication standards (communication protocol) exist
 - Both the Transceiver and Receivers must use the same standard

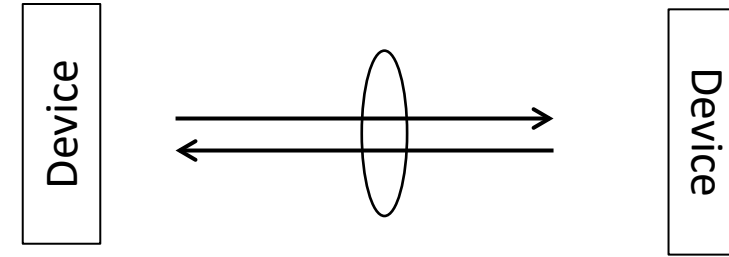
*Communication (from Latin "**communis**", meaning to **share**)
the activity of conveying information through the exchange of thoughts, messages,
or information, as by speech, visuals, signals, writing, or behavior.*

Communication

- Communication techniques
 - I/O-communication
 - Digital 24VDC signals
 - Analog signal 4-20mA, 0-10V..
 - Serial communication (Synchronous, Asynchronous)
 - Parallel communication
 - Fieldbuses
 - Devicenet, Profibus, EtherCAT,....
 - Ethernet
 - TCP-IP, Socket, FTP, UDP, OPC-UA, ModBusTCP,
 - Radio
 - Bluetooth, ZigBee,
 - Fiber

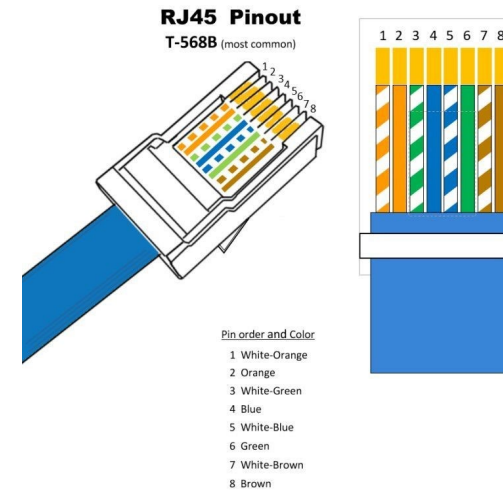
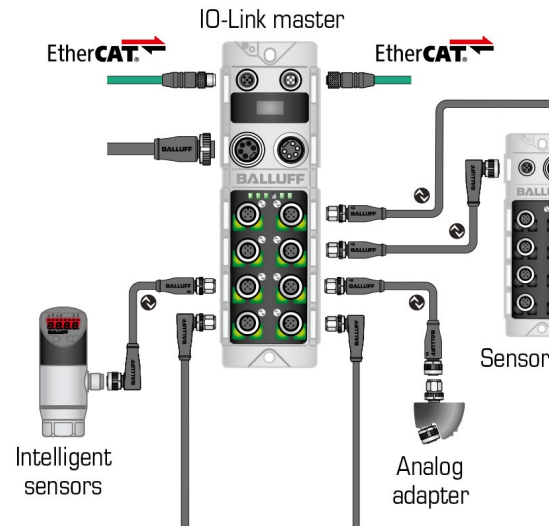
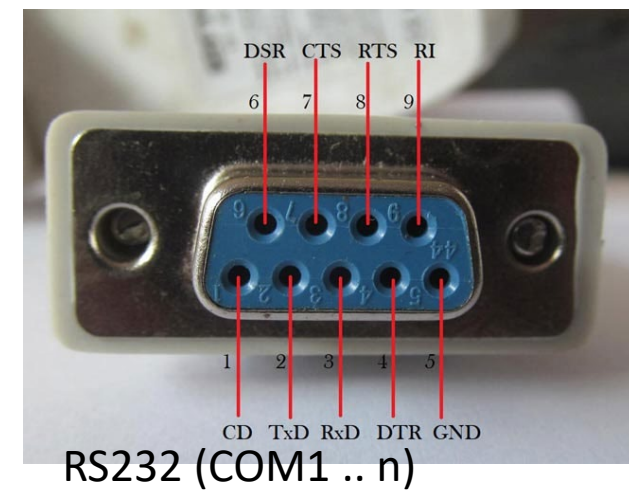
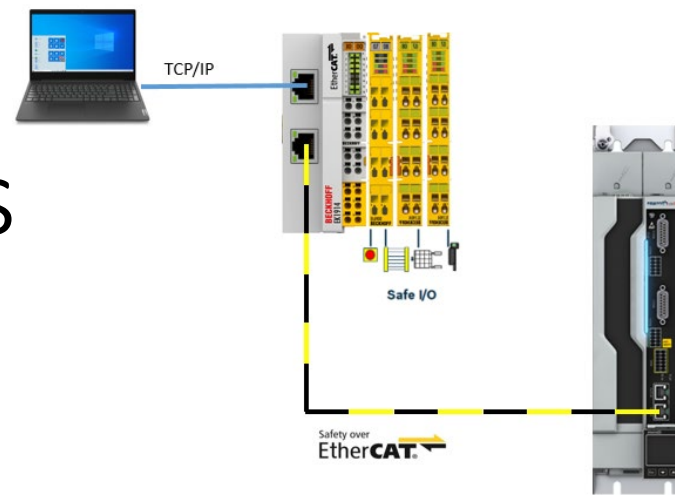


Interface and protocol



- A standardized way of “talking” both electrical and logical
- **Interface**
 - Physical connection that establishes a connection (hardware)
 - Signals (hardware) connection to Variables (software)
- **Protocol**
 - Methods and rules to achieve communication (software)

Physical Interfaces



OSI model

Layer	Function
7. Application	High-level APIs, including resource sharing, remote file access - API interface
6. Presentation	Translation of data between a networking service and an application; including character encoding, data compression and encryption/decryption - Convert and encryption
5. Session	Managing communication sessions, i.e. continuous exchange of information in the form of multiple back-and-forth transmissions between two nodes - Coordinate and Sync.
4. Transport	Reliable transmission of data segments between points on a network, including segmentation, acknowledgement and multiplexing - Divide to IP packets
3. Network	Structuring and managing a multi-node network, including addressing, routing and traffic control - Routing
2. Data link	Reliable transmission of data frames between two nodes connected by a physical layer - Error correction
1. Physical	Transmission and reception of raw bit streams over a physical medium - Electrical hardware

Protocol

- A set of formal rules describing how to transmit or exchange data
- Syntax, semantics, and synchronization of communication
- Rules and formats for exchanging messages in or between systems
- May include signaling, authentication, and error detection/correction capabilities
- Can be implemented in hardware, software, or both

Standard Protocols: Industrial

- EtherCAT
- ProfibusDP
- ProfiNET
- PowerLink
- ModBus
- CC-Link
- DeviceNet
- InterbusS
- CANOpen
- I/O-link
- ASI
- OPC-UA
- Ethernet/IP
- ADS

7.Application

DHCP

DNS

FTP

HTTP

MQTT

SNTP

SSH

Telnet

TLS/SSL

4. Transport

TCP

UDP

3. Network

IP

1. Physical

EtherNet

CAN

RS232

RS485

Protocol strategies

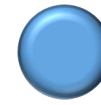
- **Demands for Real-Time V.S. data volume**
- Triggered events
 - Multi-master system, transmitting a message when there is something new to tell
- Polling systems
 - Continuous information exchange
 - Master informs Slaves that respond with an answer
- Summation frames
 - A train of information circulating between nodes of the system
- Handshake and error handling, level of fail-safe

Topology

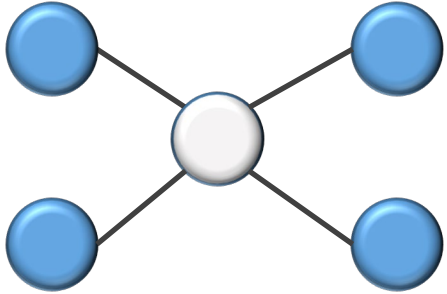
Hub



Nodes

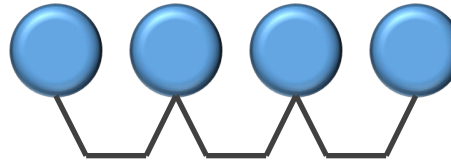


Star network



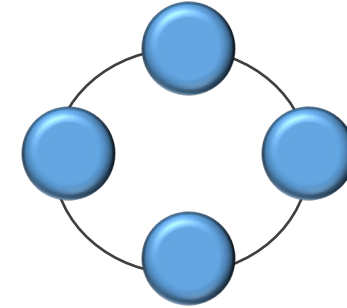
- + High fail safety
- + Less collision
- + Easy to connect disconnect units during operation
- Many cables
- All traffic through a router

Bus network



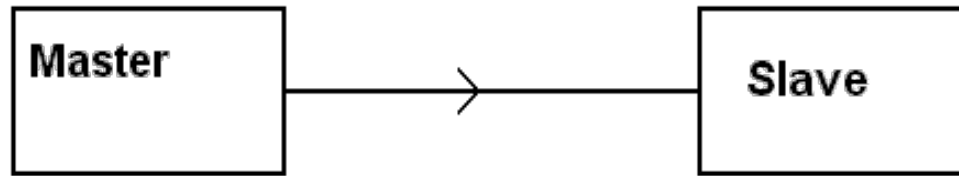
- + Easy installation
- + Easy to rebuild
- + Less cable
- Requires advanced rules for communication
- Slow when much traffic
- Everything will fail if a communication wire breaks

Ring network

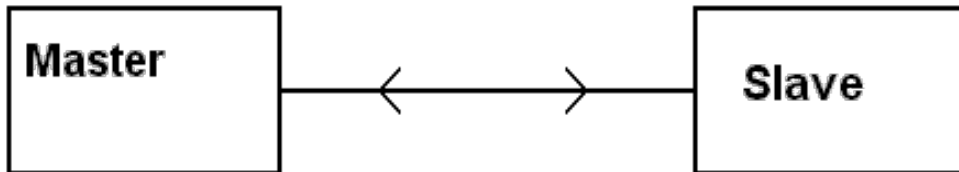


- + High fail safety
- + Can work even if a wire breaks
- Hard to rebuild
- All traffic goes through all nodes

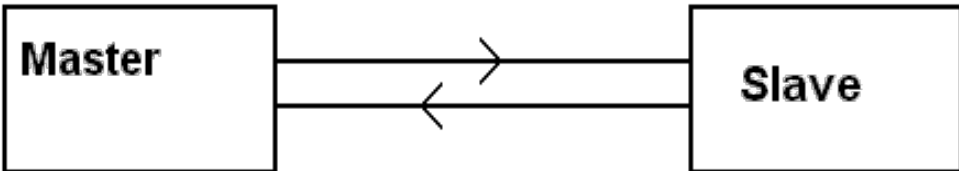
Communication directions



- **Simplex:**
Communication in one direction



- **Half duplex:**
Communication in two directions
but one-by-one

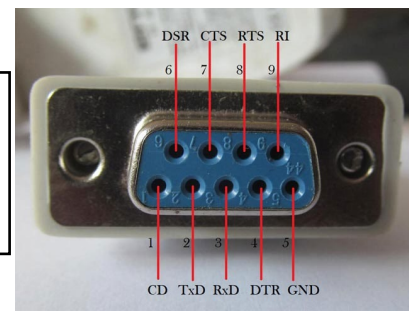
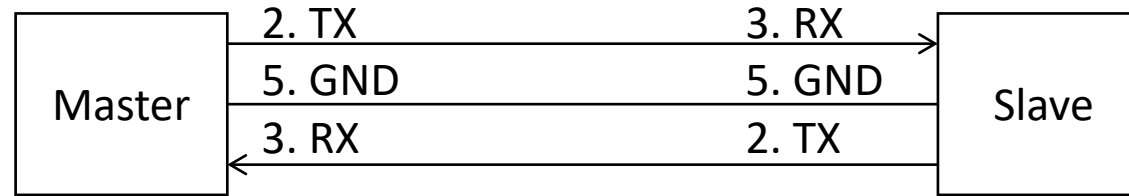


- **Full duplex:**
Communication in two directions
simultaneously

Electrical interface standards for Serial Communication

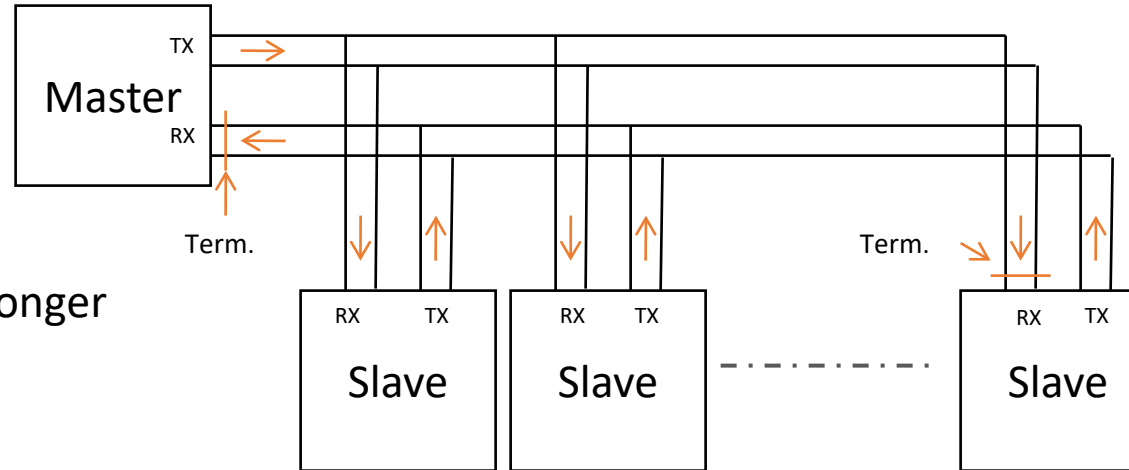
RS 232:

Full duplex, Single-ended
Introduced 1962
Max 15 meter
Named COM1, COM2, .. In a Windows computer

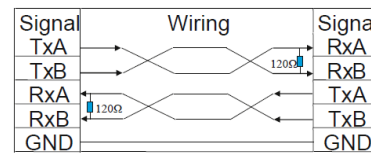


RS 422:

Full duplex, Multidrop
One driver 10 receivers
Often used for potential free
RS232 communication over longer
distances (via modem)

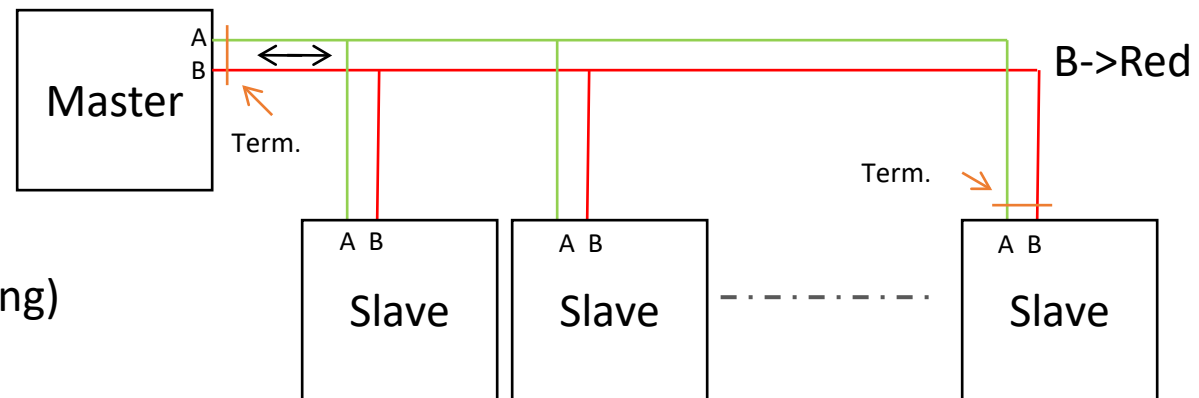


Tx Transmit data
Rx Receive data



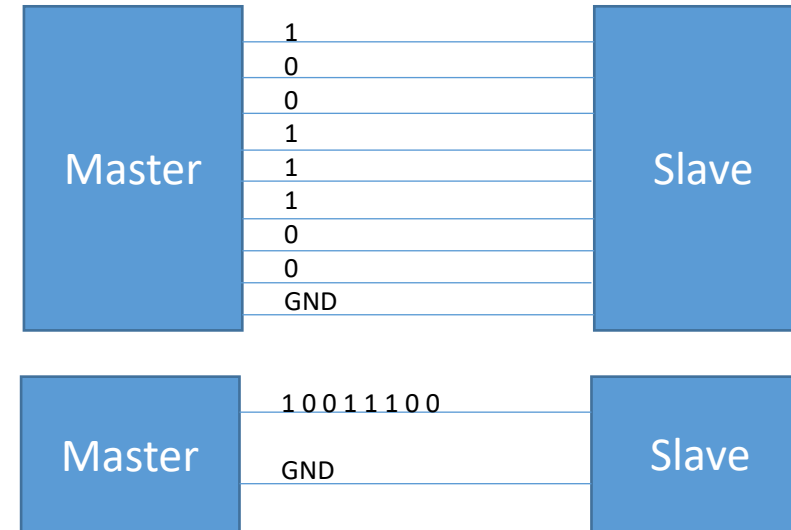
RS 485:

Half duplex, Multidrop:
One driver 32 receivers
Used by Profibus DP
The Bread rule (B to Red wiring)



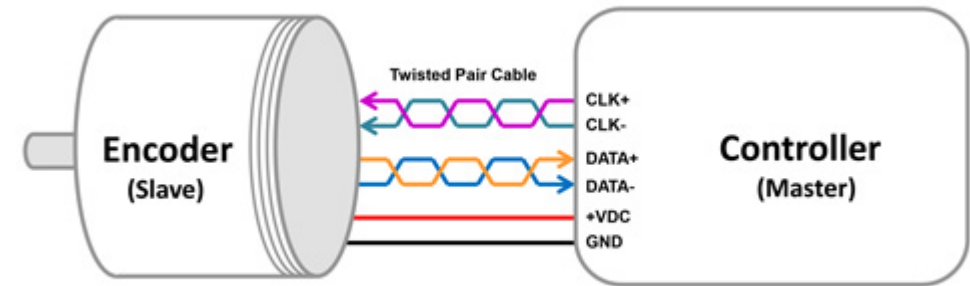
Parallel / Serial communication

- Parallel communication
 - 8-bits (byte)
 - 16-bits (word)
 - 32-bits (double word)
 - 64-bits (quad word/long word)
- Serial communication
 - Two vires
- Serial communication is mainly used due to cheap solutions and hi-speed technique



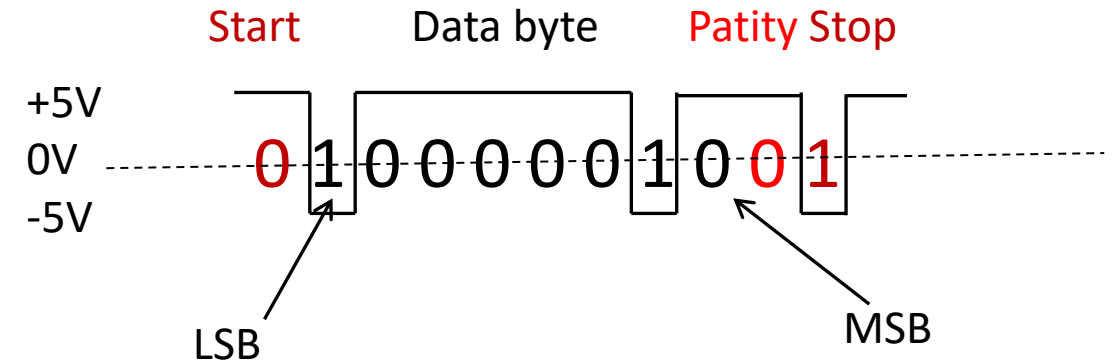
Synchronous communication

- Synchrony Serial Interface (SSI)
- Master and slave are always synchronized (Real-time)
- Depended on a common clock (CLK)
- Not common today, due to high hardware costs and extra wiring for the clock signal
- Might be hard to maintain if close to execution time-space
 - Relatively short communication distances



Asynchronous Communication

- No common clock
- Synchronization via
 - Start and stop bit(s)
 - Parity bit for fail detection
 - Start of text (telegram) STX
 - End of text (telegram) ETX
- Needs an exact clock on both ends
 - Standard Baud rates: 4800, **9600**, 19200, 28800, 38400, 57600, 76800, **115200**
 - Higher baud rate -> shorter cable



- (LSB) = Least Significant Bit
- (MSB) = Most Significant Bit
- NRZ Non-Return to Zero principle
- Each byte transmits in sequences of 11 bits: start bit, data byte, stop bit and parity bit.

7-bits ASCII Table

Dec	Hx	Oct	Char	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr	Dec	Hx	Oct	Html	Chr
0	0	000	NUL (null)	32	20	040	 	Space	64	40	100	@	@	96	60	140	`	`
1	1	001	SOH (start of heading)	33	21	041	!	!	65	41	101	A	A	97	61	141	a	a
2	2	002	STX (start of text)	34	22	042	"	"	66	42	102	B	B	98	62	142	b	b
3	3	003	ETX (end of text)	35	23	043	#	#	67	43	103	C	C	99	63	143	c	c
4	4	004	EOT (end of transmission)	36	24	044	$	\$	68	44	104	D	D	100	64	144	d	d
5	5	005	ENQ (enquiry)	37	25	045	%	%	69	45	105	E	E	101	65	145	e	e
6	6	006	ACK (acknowledge)	38	26	046	&	&	70	46	106	F	F	102	66	146	f	f
7	7	007	BEL (bell)	39	27	047	'	'	71	47	107	G	G	103	67	147	g	g
8	8	010	BS (backspace)	40	28	050	(	(	72	48	110	H	H	104	68	150	h	h
9	9	011	TAB (horizontal tab)	41	29	051	)	)	73	49	111	I	I	105	69	151	i	i
10	A	012	LF (NL line feed, new line)	42	2A	052	*	*	74	4A	112	J	J	106	6A	152	j	j
11	B	013	VT (vertical tab)	43	2B	053	+	+	75	4B	113	K	K	107	6B	153	k	k
12	C	014	FF (NP form feed, new page)	44	2C	054	,	,	76	4C	114	L	L	108	6C	154	l	l
13	D	015	CR (carriage return)	45	2D	055	-	-	77	4D	115	M	M	109	6D	155	m	m
14	E	016	SO (shift out)	46	2E	056	.	.	78	4E	116	N	N	110	6E	156	n	n
15	F	017	SI (shift in)	47	2F	057	/	/	79	4F	117	O	O	111	6F	157	o	o
16	10	020	DLE (data link escape)	48	30	060	0	0	80	50	120	P	P	112	70	160	p	p
17	11	021	DC1 (device control 1)	49	31	061	1	1	81	51	121	Q	Q	113	71	161	q	q
18	12	022	DC2 (device control 2)	50	32	062	2	2	82	52	122	R	R	114	72	162	r	r
19	13	023	DC3 (device control 3)	51	33	063	3	3	83	53	123	S	S	115	73	163	s	s
20	14	024	DC4 (device control 4)	52	34	064	4	4	84	54	124	T	T	116	74	164	t	t
21	15	025	NAK (negative acknowledge)	53	35	065	5	5	85	55	125	U	U	117	75	165	u	u
22	16	026	SYN (synchronous idle)	54	36	066	6	6	86	56	126	V	V	118	76	166	v	v
23	17	027	ETB (end of trans. block)	55	37	067	7	7	87	57	127	W	W	119	77	167	w	w
24	18	030	CAN (cancel)	56	38	070	8	8	88	58	130	X	X	120	78	170	x	x
25	19	031	EM (end of medium)	57	39	071	9	9	89	59	131	Y	Y	121	79	171	y	y
26	1A	032	SUB (substitute)	58	3A	072	:	:	90	5A	132	Z	Z	122	7A	172	z	z
27	1B	033	ESC (escape)	59	3B	073	;	:	91	5B	133	[	[	123	7B	173	{	{
28	1C	034	FS (file separator)	60	3C	074	<	<	92	5C	134	\	\	124	7C	174	|	
29	1D	035	GS (group separator)	61	3D	075	=	=	93	5D	135	]	]	125	7D	175	}	}
30	1E	036	RS (record separator)	62	3E	076	>	>	94	5E	136	^	^	126	7E	176	~	~
31	1F	037	US (unit separator)	63	3F	077	?	?	95	5F	137	_	_	127	7F	177		DEL

Example of an communication protocol

ASCII Table	
STX	02
ETX	03
!	33
0	48
1	49
2	50
A	65
B	66
C	67

Start Bit Data byte Parity Stop

0 1 0 0 0 0 0 1 0 0 1

STX	Address	Number of data byte	DATA	DATA	DATA	Checksum XOR	ETX
02	49	51	65	66	67	66	03

Overhead

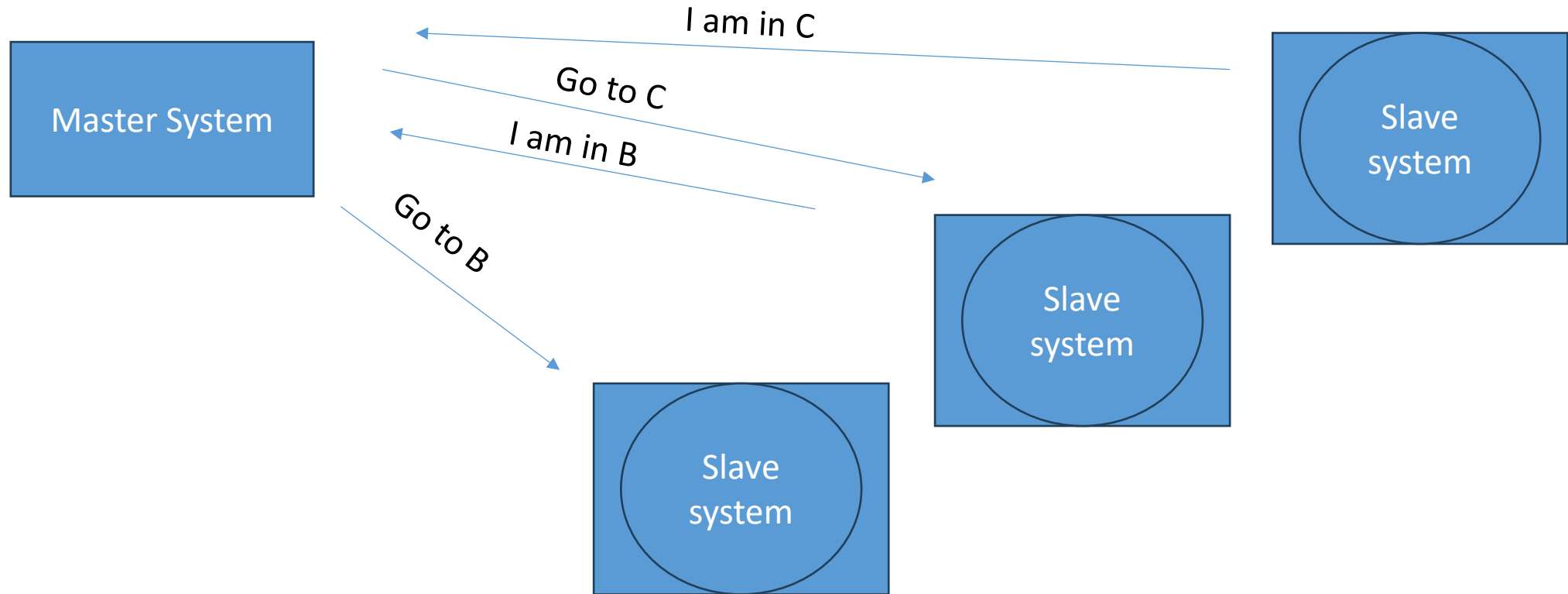
Overhead



Implement a handshake procedure for communication between two not-synchronized computer systems

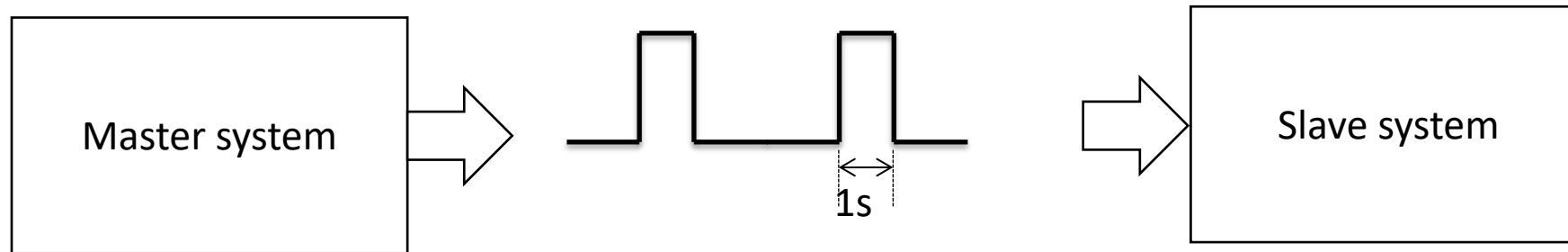
- The transmitting system must wait for the answer and acknowledge
- The receiving system must wait for the acknowledgment

Handshake to synchronize two systems



Enable trigger signal

- Easy to implement
- Usually **too simple for safe functionality**
- No safe restart
 - Requires some time perception or knowledge about the process
 - No feedback



Synchronization between a PLC
and a robot is not standardized
and must be implemented by
code
both in the PLC and robot

Example of communication via digital I/O

Hardware Setup (Interfaces)



QX0.1

QX0.2

QX0.3

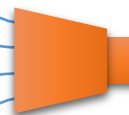
QX0.4

IX0.1

IX0.2

IX0.3

IX0.4



I0.1

I0.2

I0.3

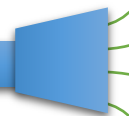
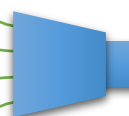
I0.4

O0.1

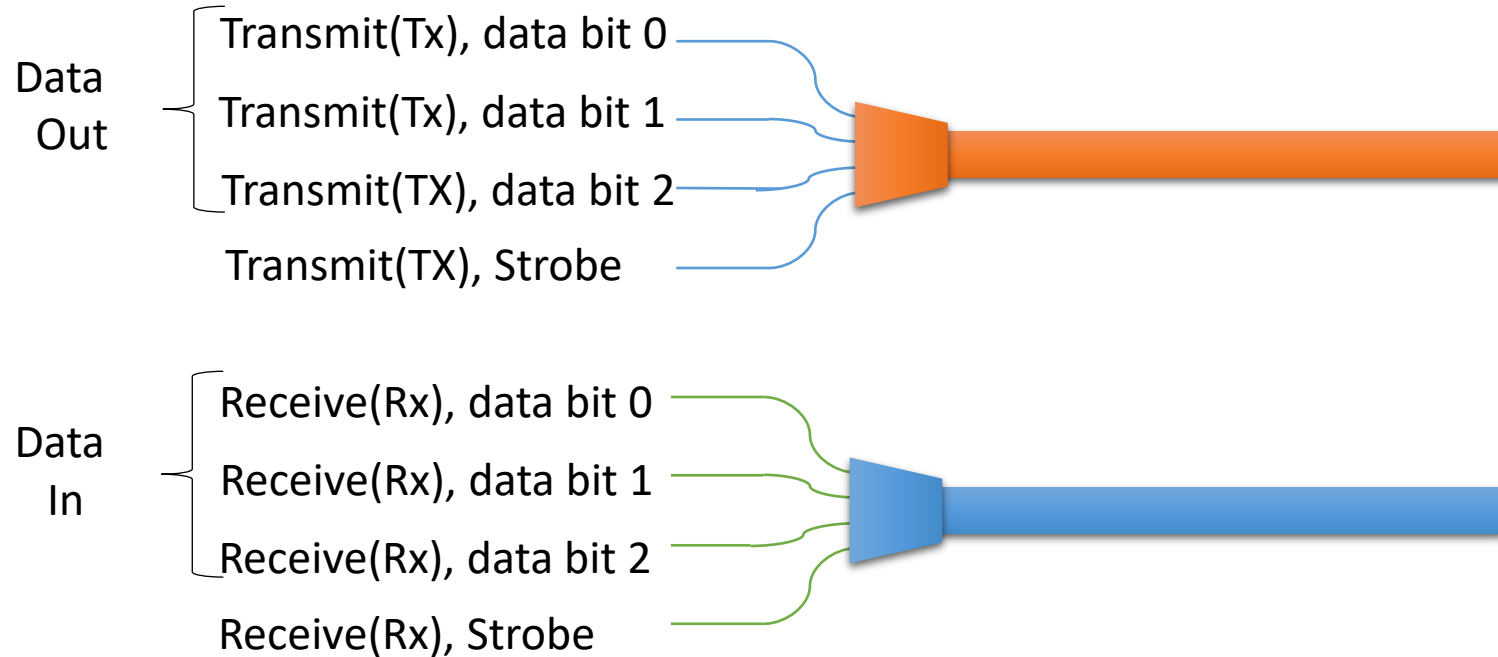
O0.2

O0.3

O0.4



PLC Logical Interface + Strobe



Integer representation by bits

- n bits can be combined in 2^n combinations
- Example: 3 bits gives 8 combinations
- 4th bit is used as a strobe

Bits	Integer
0 0 0	0
0 0 1	1
0 1 0	2
0 1 1	3
1 0 0	4
1 0 1	5
1 1 0	6
1 1 1	7
Strobe	8

Example: a 3-bit protocol in-between a PLC and Robot

- PLC is the master sending commands and the Robot is the slave and answer

Command	Integer
	0
Start Prg1	1
Start Prg2	2
Go to Home	3
	4
	5
	6
	7
Strobe	8

Implementation

PROGRAM Communication

VAR

```
// Transmitting (Tx) outgoing signals to the robot
qxTxDataBit0 AT %Q*: BOOL; // Data bit 1 value 1
qxTxDataBit1 AT %Q*: BOOL; // Data bit 2 value 2
qxTxDataBit2 AT %Q*: BOOL; // Data bit 3 value 4
qxTxStrobe AT %Q*: BOOL; // Strobe signal
```

```
// Receiving (Rx) incoming signals from the robot
ixRxDataBit0 AT %I*: BOOL; // Data bit 1 value 1
ixRxDataBit1 AT %I*: BOOL; // Data bit 2 value 2
ixRxDataBit2 AT %I*: BOOL; // Data bit 3 value 4
ixRxStrobe AT %I*: BOOL; // Strobe signal
```

```
miState: INT; // State (sequence) variable
mxGotoHome: BOOL; // Signals from the control program
mxStartPrg1: BOOL;
mxStartPrg2: BOOL;
mxRobotInHomePos: BOOL; // Signals to the control program
mxPrg1Done: BOOL;
mxPrg2Done: BOOL;
```

END_VAR

OBS! No error handling is implemented here.
What happens if the robot is not ready to run or
does not reach the home pos?
(At least a time-out function or reset is needed)

```
// Communication sequence PLC to Robot signaling
CASE miState OF
```

```
0: // Clear all Tx variables
    qxTxDataBit0 := FALSE;
    qxTxDataBit1 := FALSE;
    qxTxDataBit2 := FALSE;
    qxTxStrobe := FALSE;
    // Wait until the robot cleared all signals
    IF NOT ixRxDataBit0 AND NOT ixRxDataBit1 AND NOT ixRxDataBit2 AND NOT ixRxStrobe THEN
        miState := 10;
    END_IF

10: // Wait for an order from the control program
    IF mxGotoHome THEN
        miState := 100;
    END_IF
    IF mxStartPrg1 THEN
        miState := 200;
    END_IF
    IF mxStartPrg2 THEN
        miState := 300;
    END_IF

100: // Send the message (Value 3) go to home
    qxTxDataBit0 := TRUE;
    qxTxDataBit1 := TRUE;
    qxTxDataBit2 := FALSE;
    qxTxStrobe := FALSE;
    miState := 110;

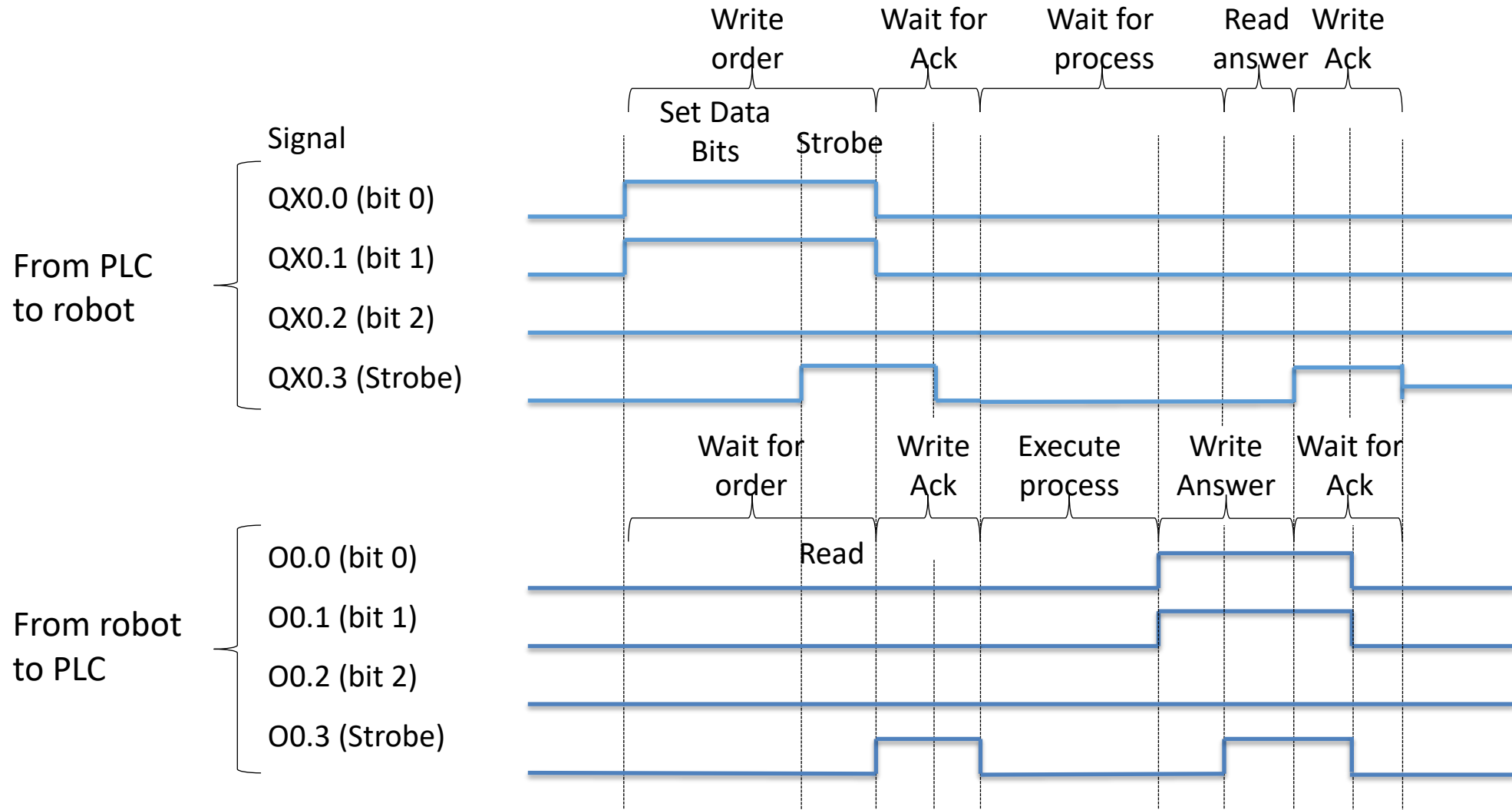
110: // Send the Strobe
    qxTxStrobe := TRUE;
    miState := 120;

120: // Wait for the answer that the robot reached the home position
    IF ixRxStrobe THEN
        miState := 130;
    END_IF

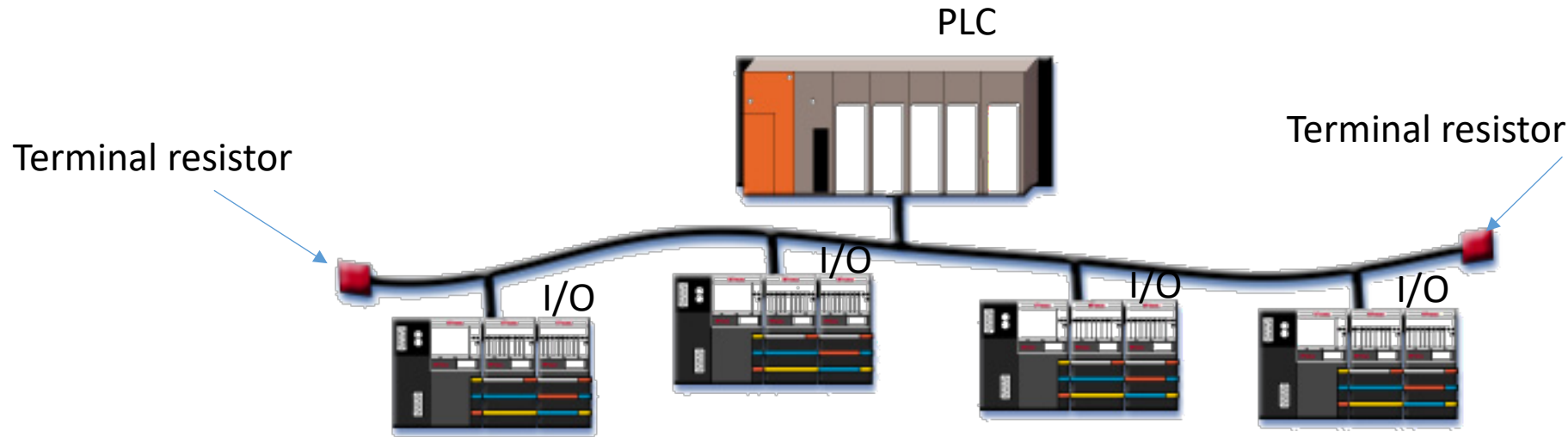
130: // Read the answer (value 3) and tell the control program that the robot is in home
    IF ixRxDataBit0 AND ixRxDataBit1 AND NOT ixRxDataBit2 THEN
        mxRobotInHomePos := TRUE;
        mxPrg1Done := FALSE;
        mxPrg2Done := FALSE;
    END_IF
    miState := 0;

200: // Send the message (Value 1) run program 1
```

Signal pattern Send Go to Home (2#011)



What is a field bus?

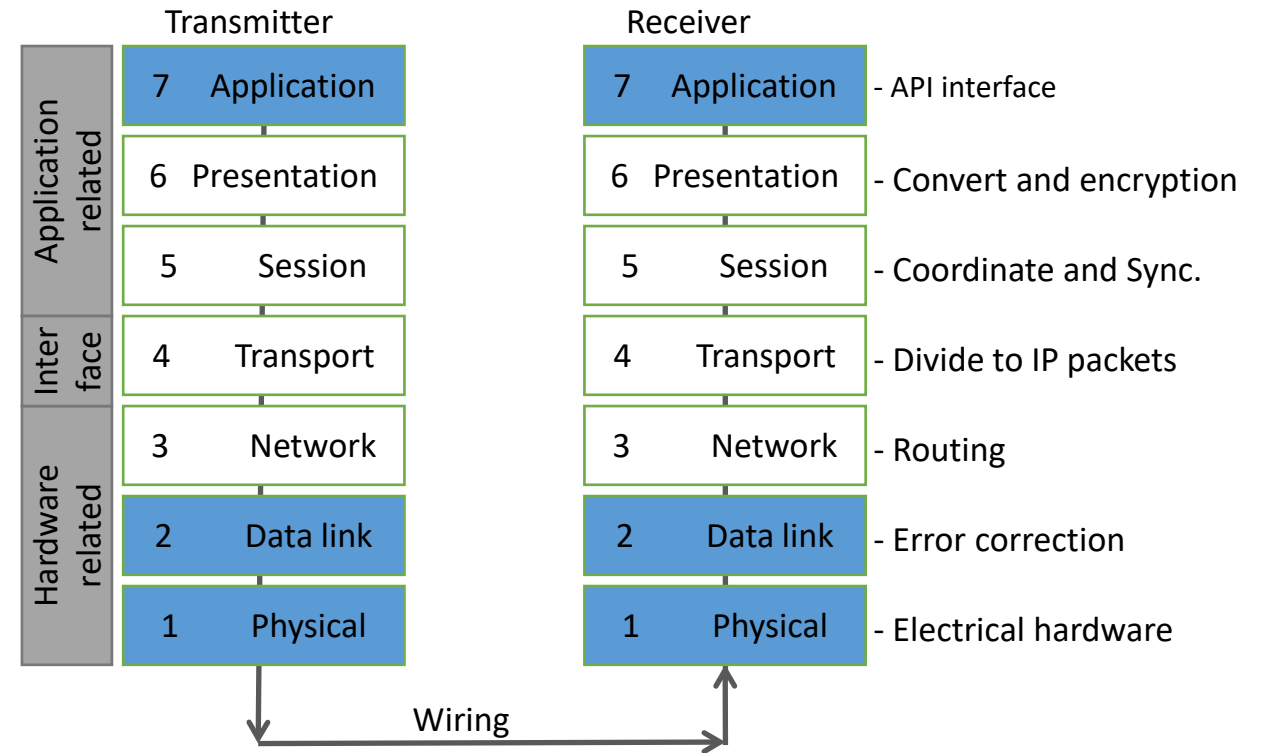


- Communication between PLC and distributed units
- Brand-dependent field buses were developed and used during the 1980s
- Open field bus standards were released at the end of the 1980s
 - Enable communication between units of different brands

The OSI model

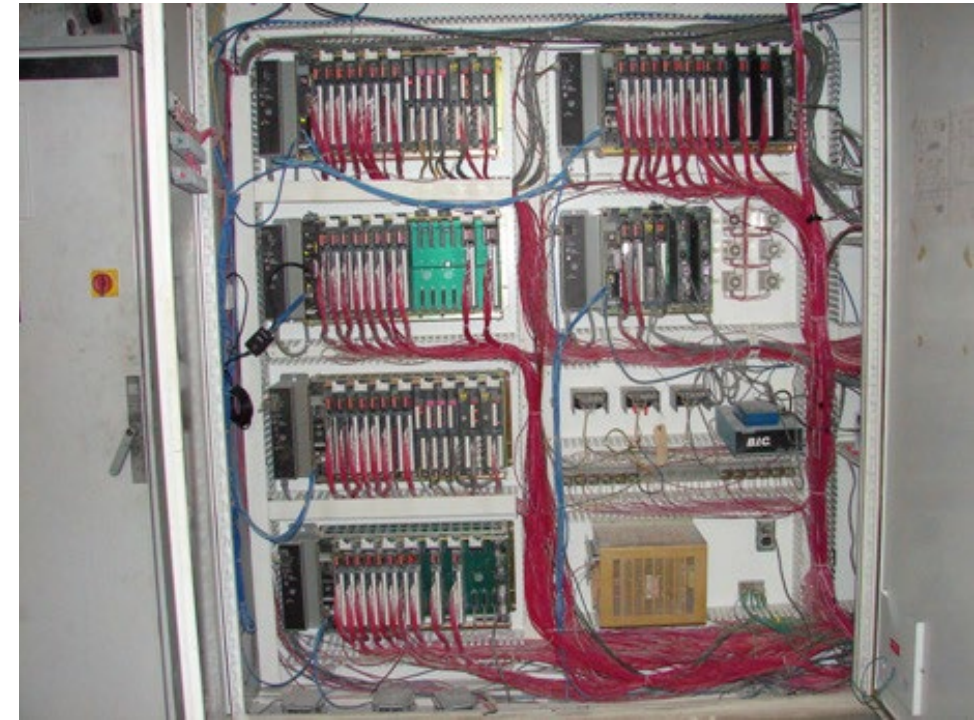
(Open System Interconnection)

- Only layer 1,2 & 7 are implemented in the field buses
- That's why the implementation of layer 5 coordinate and synchronization (handshaking) is necessary



Why field bus?

- One cable for communication to
 - I/O-modules
 - servo-drives, frequency inverters
 - operator panels, robots
- Reduces the number of cables
- Modular construction
- Flexible, easier to rebuild
- Less electrical documentation



PLC cabinet with central I/O-units



PLC-cabinet with field bus distributed I/O

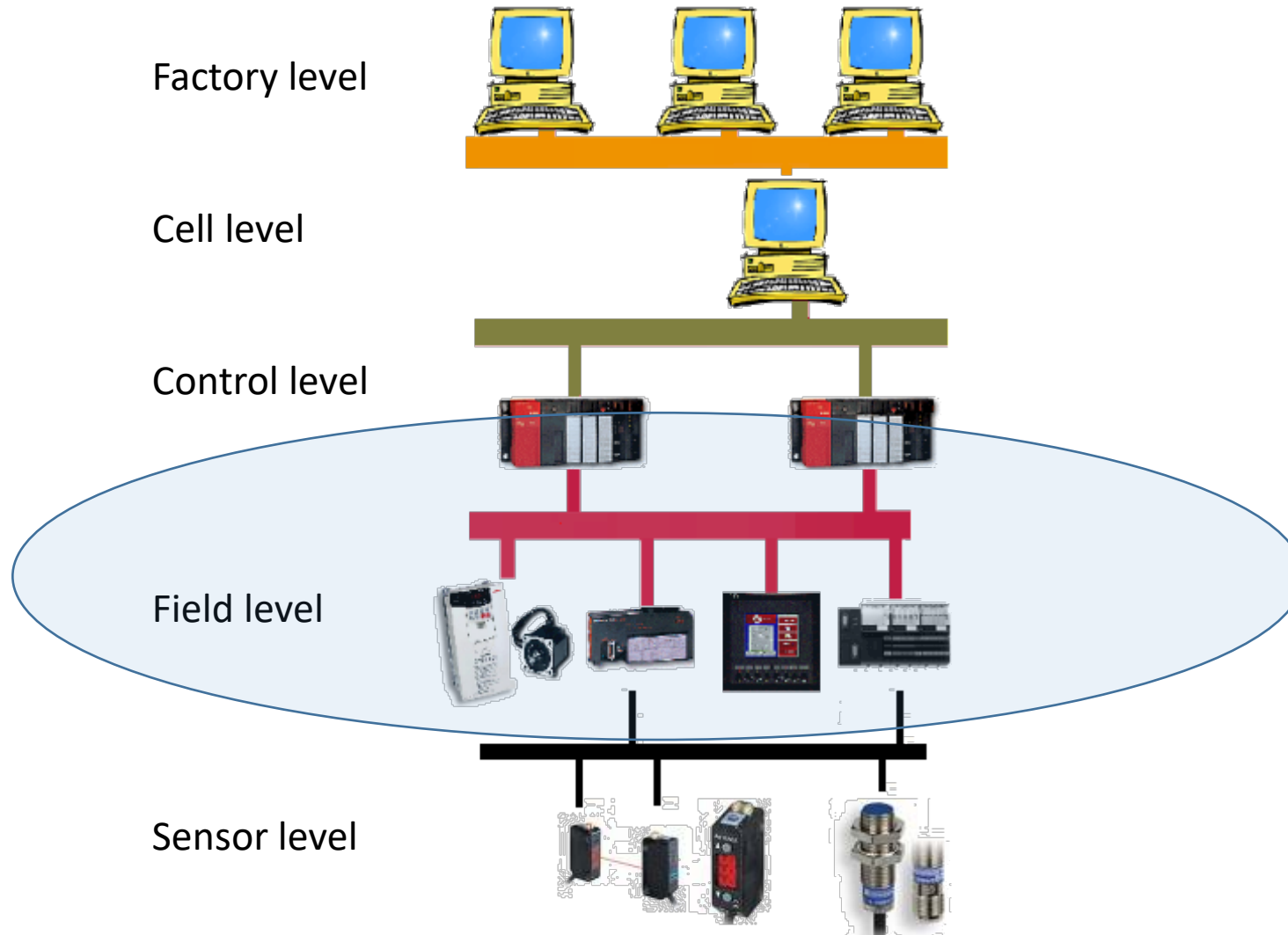
When the field bus is preferable

- Communication within the area of a robot or machine cell
- High real-time requirements and a low amount of data
- Communication to I/O-modules, motor starters, frequency inverters, servo drives, HMI, CNC, and robot controllers
- For plants that are often rebuilt

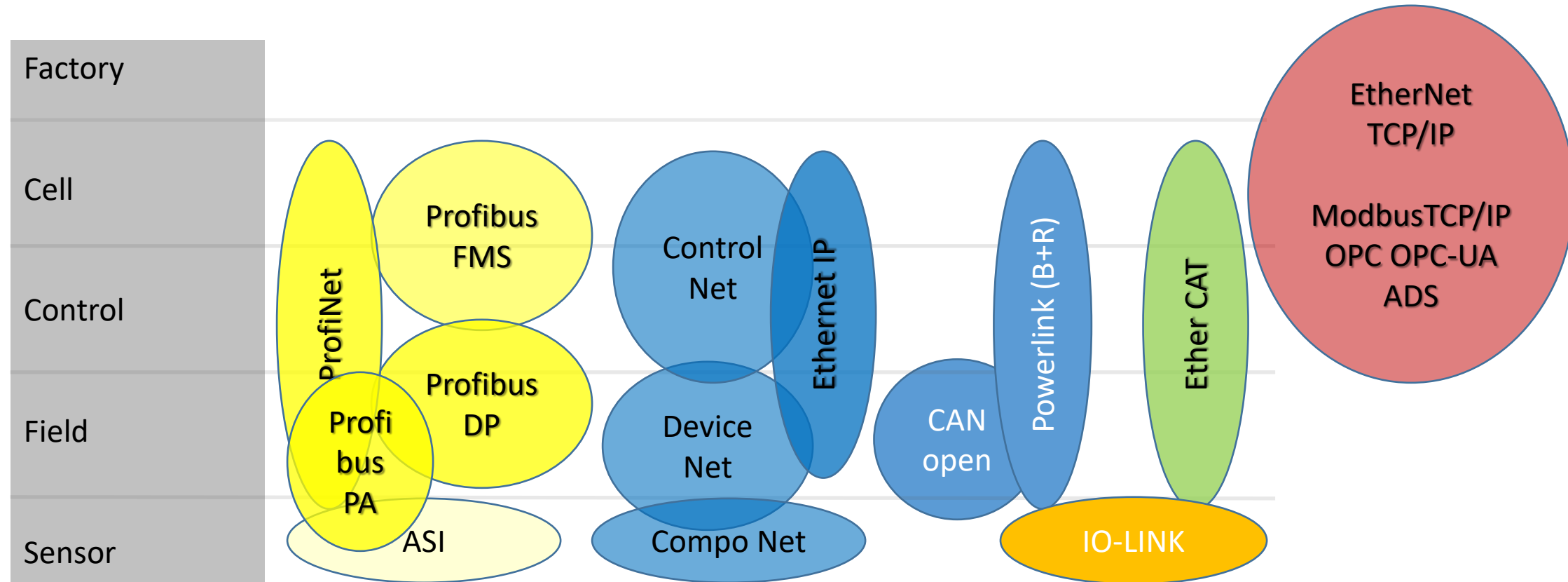
Are there any disadvantages to using a field bus

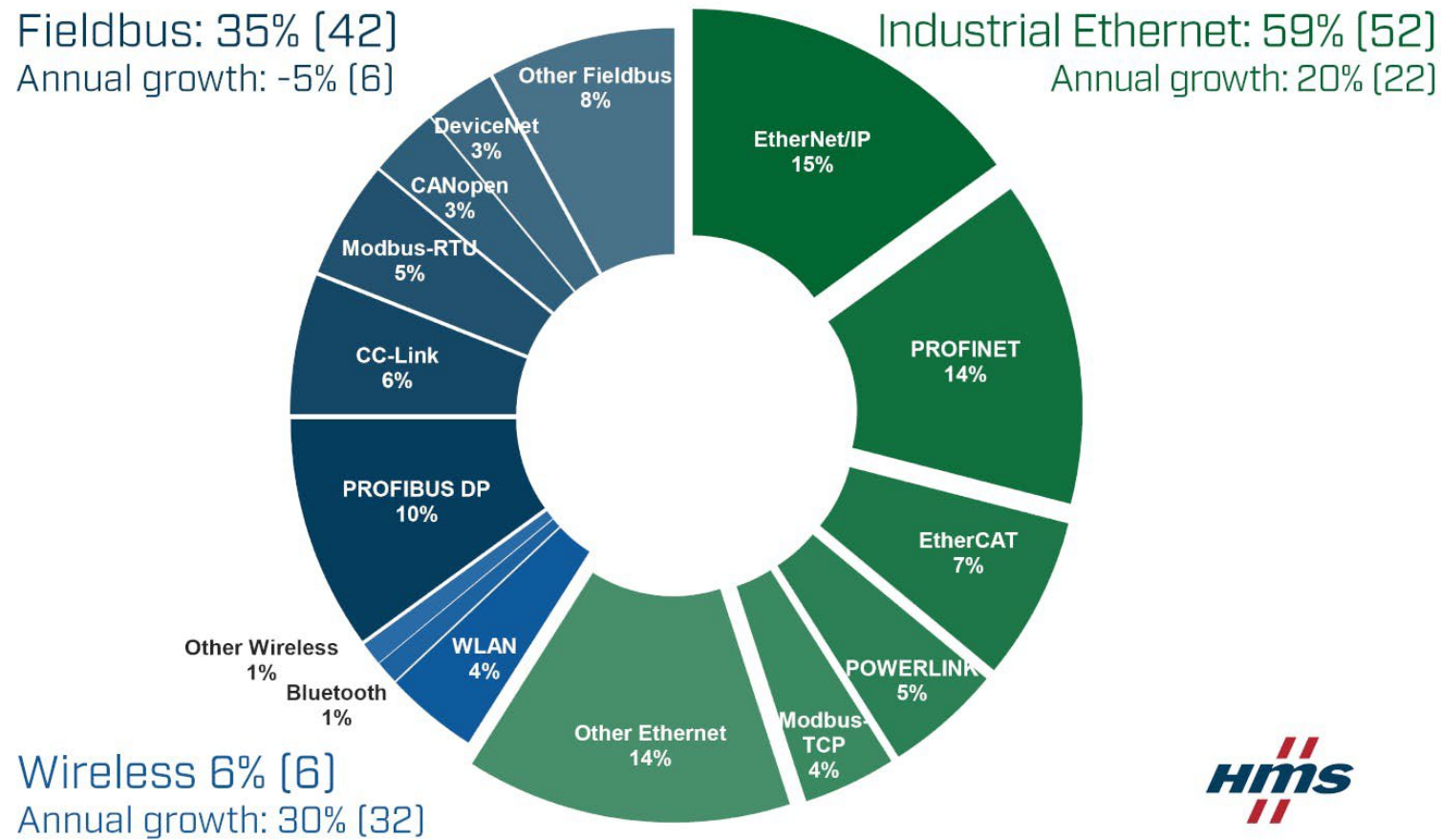
- Many different standards that are difficult to combine.
- Limited communication distances
- May not always “fail safe”
- Can sometimes be hard to configure (mix of brands)
- Difficult to troubleshoot when it malfunctions
-

Communication levels in a factory



Communication standards divided in levels





Automation World 2019

<https://www.automationworld.com/home/news/13319803/industrial-ethernet-and-wireless-growing-steadily-first-year-of-decline-for-fieldbuses-in-terms-of-new-nodes>

Communication at the automation lab

System 1	System 2	Bus
PLC	Robot controller	Profibus DP
PLC	PLC	OPC-UA, ModBus, ADS
Robot controller	I/O-modules	Device net/EthernetIP
PLC	I/O-modules	EtherCAT
Safety PLC	Safety PLC	Pluto safety bus (CAN)